

Introduction to the **TTU Climate Center** and the **National Climate Assessment**

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**CLIMATE
CENTER**



U.S. Global Change Research Program

**National Climate
Assessment**

Outline

- TTU Climate Center
 - Who we are
 - What we do
- National Climate Assessment (NCA5)
 - What is the NCA
 - NCA5 Ch. 26: Southern Great Plains



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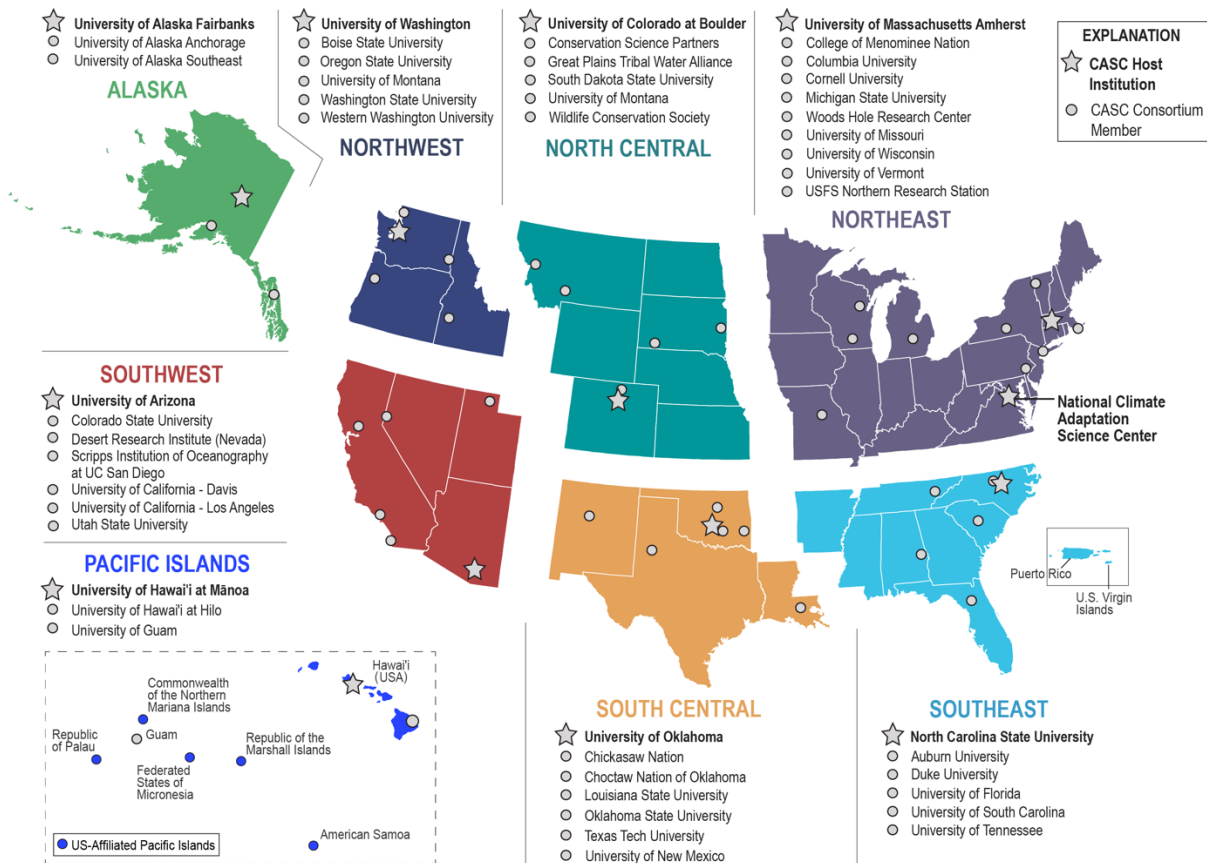
**CLIMATE
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TTU Climate Center: Who we are

- Member of USGS-funded Climate Adaptation Science Centers (CASCs)

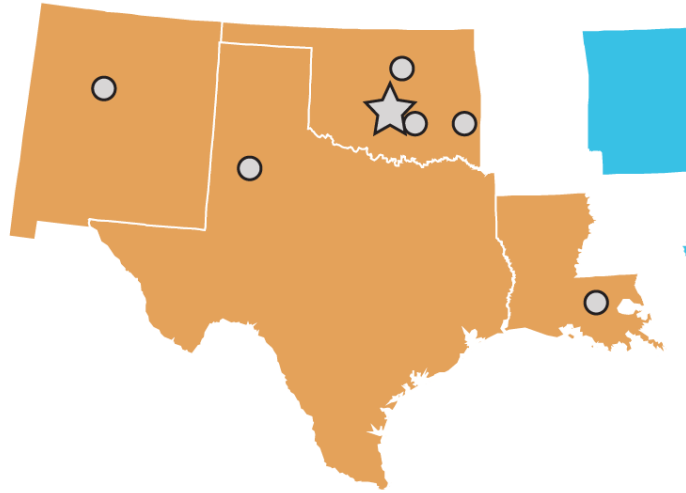
Climate Adaptation Science Center (CASC) Regions

The CASCs collaborate across boundaries to address shared ecosystems, watersheds, and landscapes



CLIMATE CENTER

TTU Climate Center: Who we are



SOUTH CENTRAL

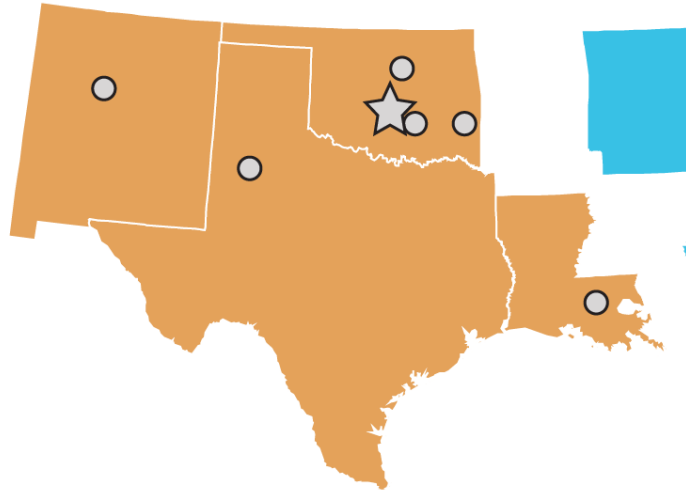
- ★ University of Oklahoma
- Chickasaw Nation
- Choctaw Nation of Oklahoma
- Louisiana State University
- Oklahoma State University
- Texas Tech University
- University of New Mexico

- Consortium institution of the South Central CASC (SC-CASC)



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TTU Climate Center: Who we are



SC-CASC:

- **Provide decision makers** with the science, tools, and information they need to address the impacts of climate variability and change
- **Transform how climate science is conducted** and applied in the south-central US
- **Support big thinking**, including multi-institutional and stakeholder-driven approaches to assessing the impact of climate on natural and cultural resources

SOUTH CENTRAL

- ★ University of Oklahoma
- Chickasaw Nation
- Choctaw Nation of Oklahoma
- Louisiana State University
- Oklahoma State University
- Texas Tech University
- University of New Mexico



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TTU Climate Center: Who we are

- >100 active members
 - Faculty, students and postdocs
- 12 TTU Colleges
- 36 TTU Departments
- Open to anyone interested in, passionate about, or doing climate-related work



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TTU Climate Center: What we do

- Serve as regional hub for climate change
- Work with community members and stakeholders
- All areas of research that intersect with climate
- Monthly research and outreach meetings
- Events
 - Informational events on and off campus
 - Science by the Glass



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Outline

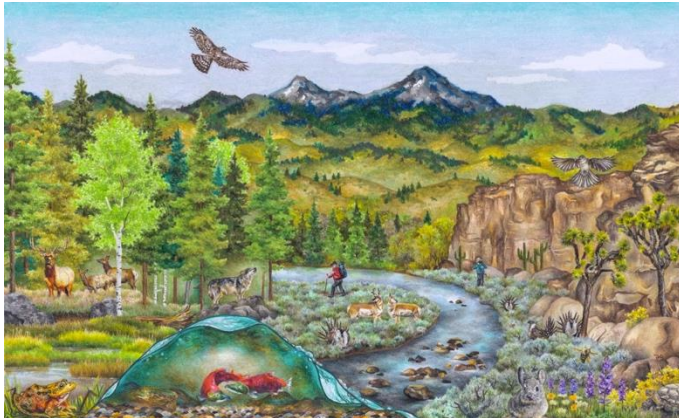
- TTU Climate Center
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U.S. Global Change Research Program
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What is the NCA?

- US government report
 - Congressionally mandated and overseen by multiple agencies
 - Delivered to congress and the president
- Climate change impacts, risks, and responses for USA
- Provides scientific foundation for decision making
- Policy relevant, but policy neutral



What is NCA? Mandate

- Result of the Global Change Research Act of 1990
- Every 4 years a report that “analyzes the effects of global change on the natural environment, agriculture, energy production and use, land and water resources, transportation, human health and welfare, human social systems, and biological diversity; and analyzes current trends in global change, both human-induced and natural, and projects major trends for the subsequent 25 to 100 years”



What is NCA? Review process

- Reviewed by:
 - Federal government experts
 - External experts
 - General public
 - External reviewers from NASEM
- Review by each group multiple times throughout the process
- All reviews responded to



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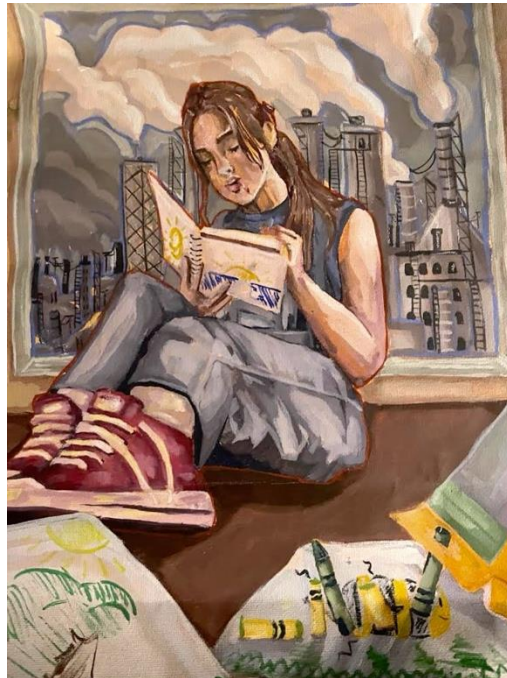
What is NCA? Audience

- Audience: decision makers
 - national, state, local, and Tribal governments, city planners, public health officials, adaptation specialists, nurses, farmers, business owners, community organizers, researchers, water utilities, ecosystem managers, educators, students, the media, and concerned individuals



What is NCA? Structure

- Each chapter has list of key messages
 - Conclusions based on authors' expert judgment and synthesis of the assessed information sources
 - Often present findings in the context of risks to natural and/or human systems



What is NCA? Confidence and Likelihood

- Calibrated terms in key messages
 - **Confidence**: based on the type, amount, quality, strength, and consistency of evidence; the skill, range, and consistency of methods to detect, evaluate, attribute, and interpret climate trends; and the degree of agreement across scientific information sources
 - **Likelihood**: based on measures of certainty expressed probabilistically



What is NCA? Content

- Four categories of chapters
 - Overview chapter (Ch. 1)
 - Physical science chapters (Ch. 2-3)
 - National topics chapters (Ch. 4-20)
 - Regional chapters (Ch. 21-30)
 - Response chapters (Ch. 31-32)



U.S. Global Change Research Program
**National Climate
Assessment**

About this Report

Guide to this Report

OVERVIEW

PHYSICAL SCIENCE

2. Climate Trends

3. Earth Systems Processes

NATIONAL TOPICS

4. Water

5. Energy

6. Land

7. Forests

8. Ecosystems

9. Coasts

10. Oceans

11. Agriculture

12. Built Environment

13. Transportation

14. Air Quality

15. Human Health

16. Indigenous Peoples

17. International

18. Complex Systems

19. Economics

20. Social Systems and Justice

REGIONS

21. Northeast

22. Southeast

23. US Caribbean

24. Midwest

25. Northern Great Plains

26. Southern Great Plains

27. Northwest

28. Southwest

29. Alaska

30. Hawai'i and US-Affiliated Pacific Islands

RESPONSES

31. Adaptation

32. Mitigation

FOCUS ON

F1. Compound Events

F2. Western Wildfires

F3. COVID-19 and Climate Change

F4. Risks to Supply Chains

F5. Blue Carbon

APPENDICES

A1. Process

A2. Information Quality

A3. Scenarios and Datasets

A4. Indicators

A5. Glossary

All Figures

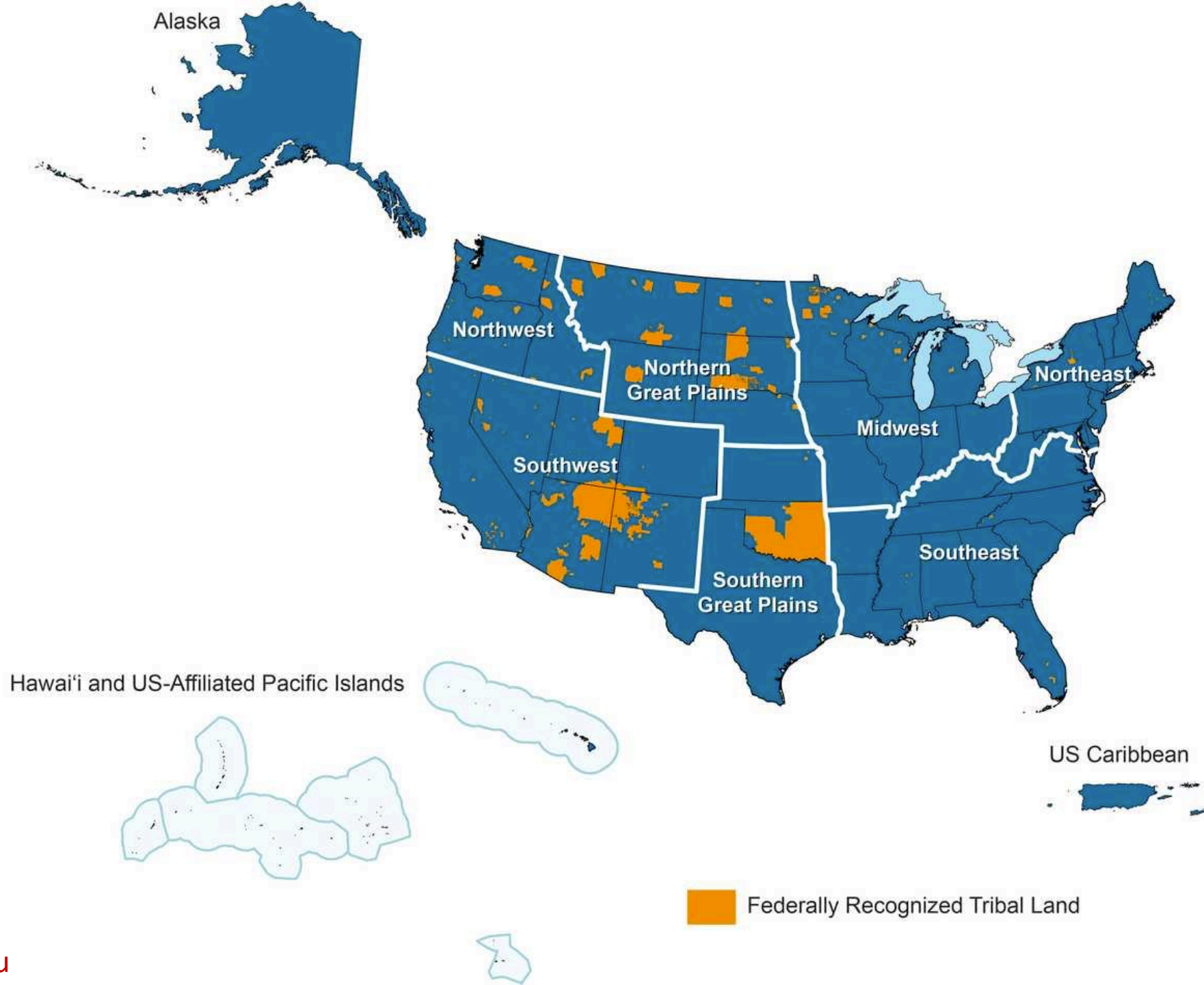
All Key Messages

What is NCA5? Content

- Four categories of chapters
 - Overview chapter (Ch. 1)
 - Physical science chapters (Ch. 2-3)
 - National topics chapters (Ch. 4-20)
 - **Regional chapters (Ch. 21-30)**
 - Response chapters (Ch. 31-32)



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FIFTH NATIONAL CLIMATE ASSESSMENT

26. Southern Great Plains



« PREVIOUS CHAPTER

 VIEW ALL FIGURES

NEXT CHAPTER »



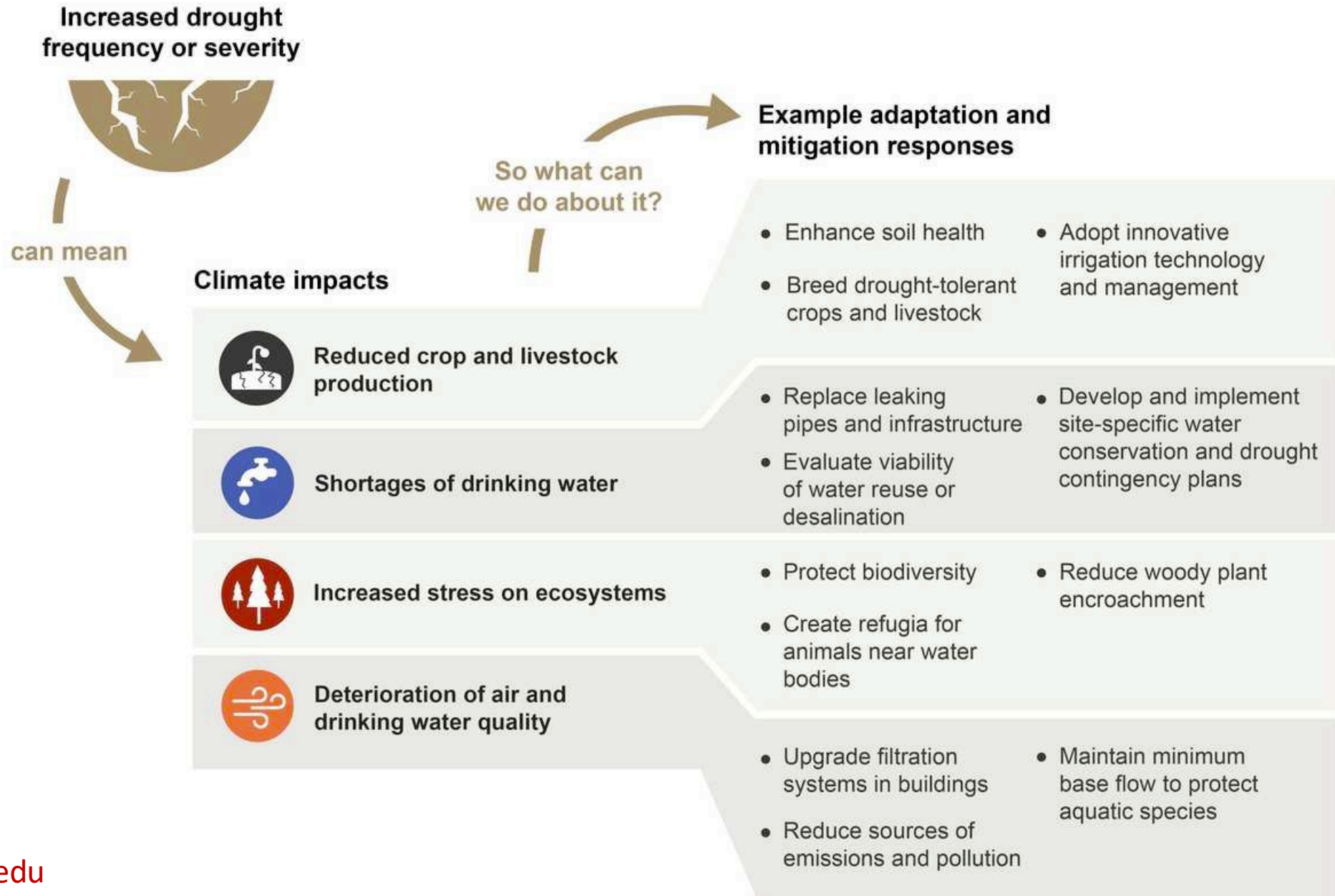
SECTIONS

Introduction

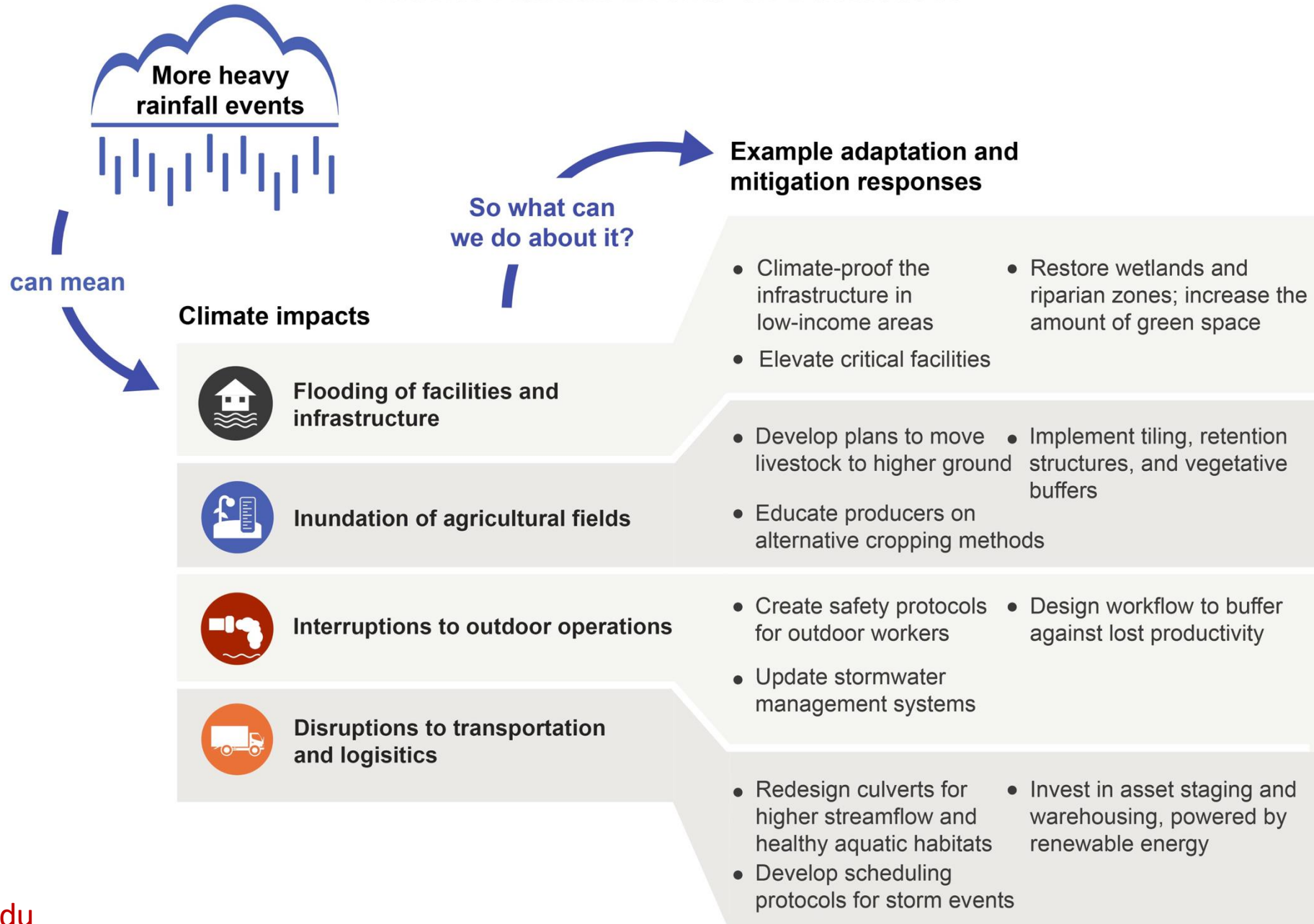
Health-related Key Messages from NCA5 SGP chapter

- Climate change is putting us at risk from climate hazards that **degrade our health and well-being** (high confidence).
- Climate change is expected to **increase heat-related illness and death** (very likely, high confidence).
- Climate change will **compound existing burdens**, with the greatest impact on people with the fewest resources to prepare and adapt (very high confidence).
- The **institutions that serve our communities have been challenged** to respond and adapt (medium confidence).

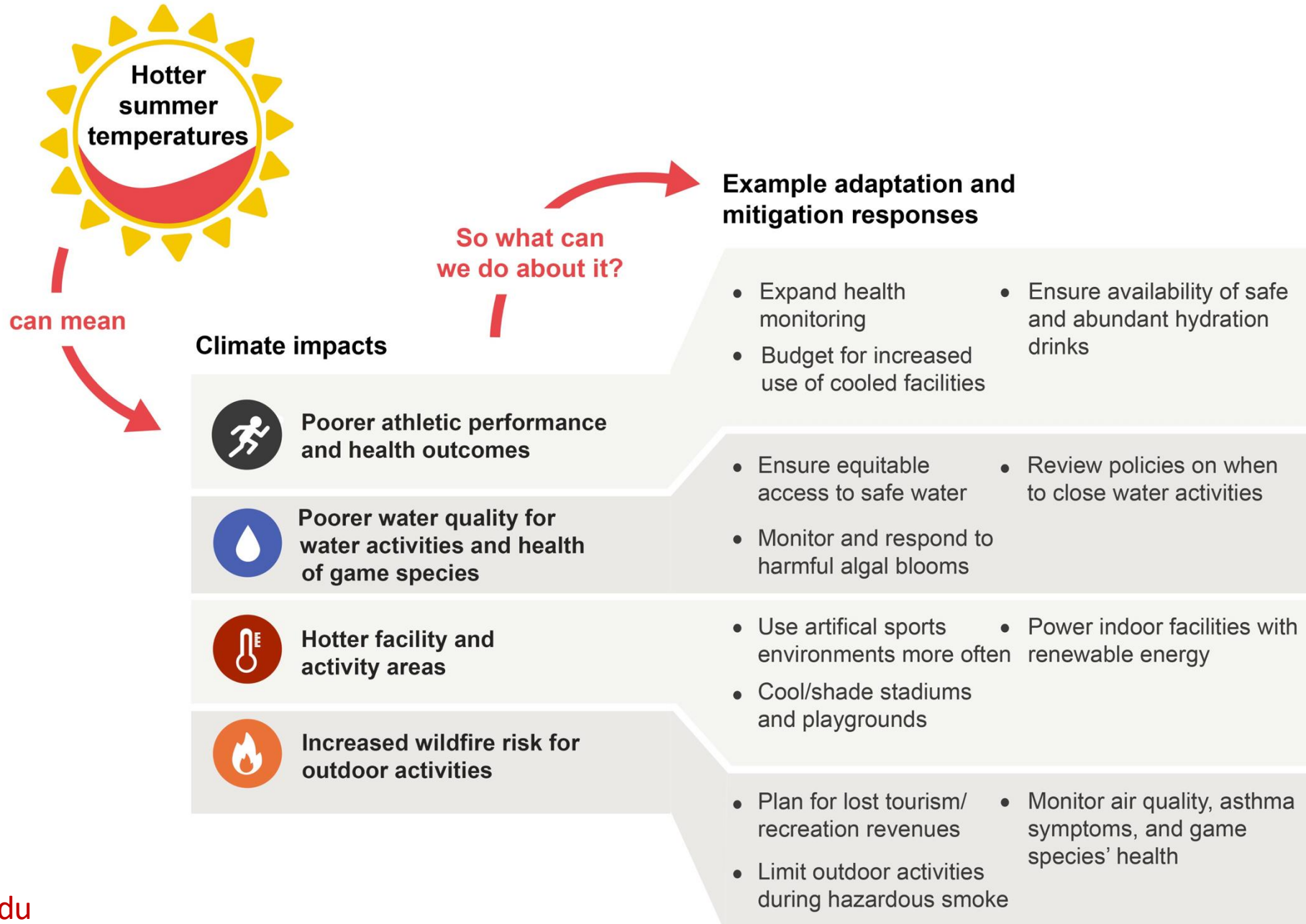
Resilience Actions to Address the Impacts of More Frequent or Severe Droughts on Communities



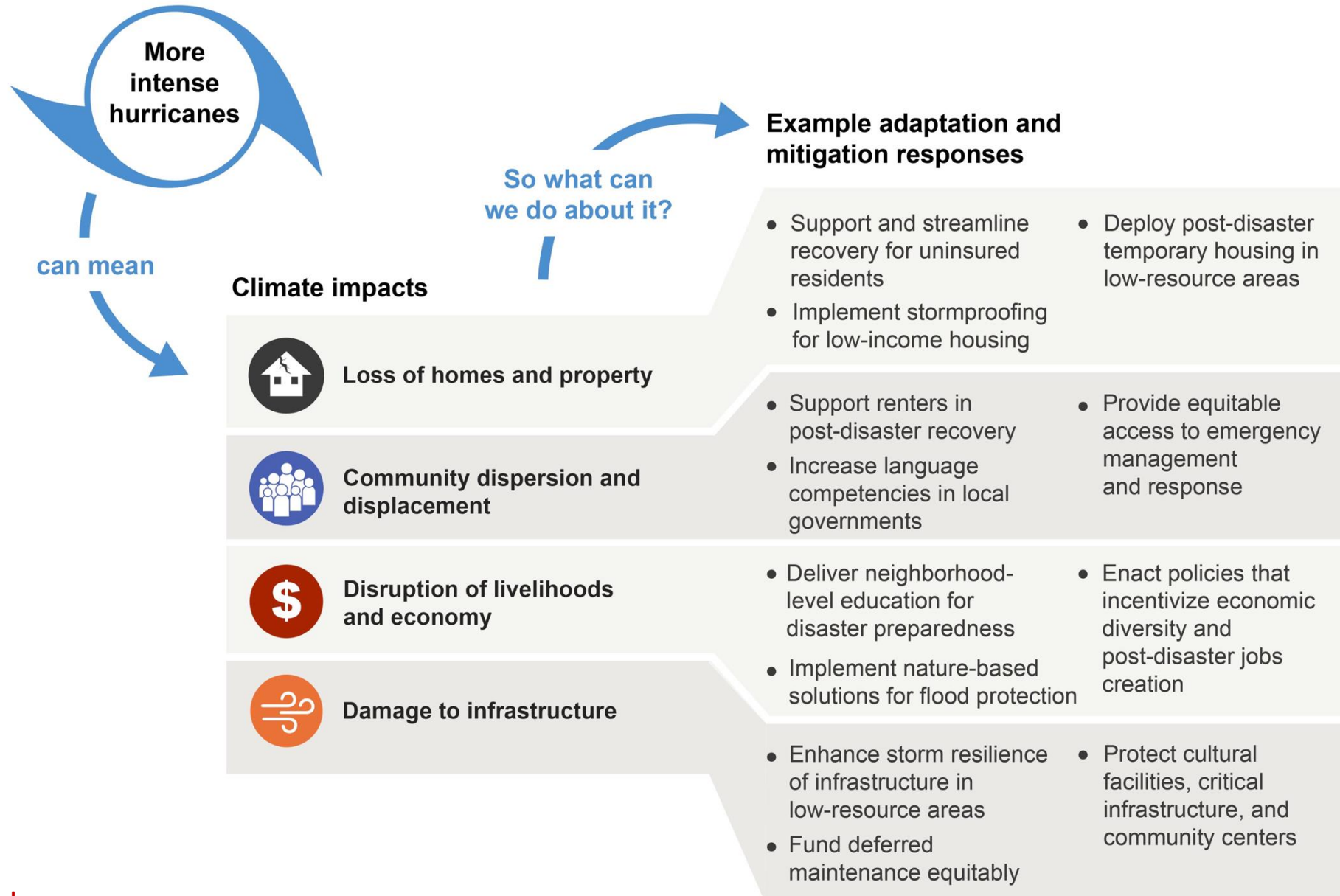
Resilience Actions to Address the Impacts of Heavier Rainfall Events on Businesses



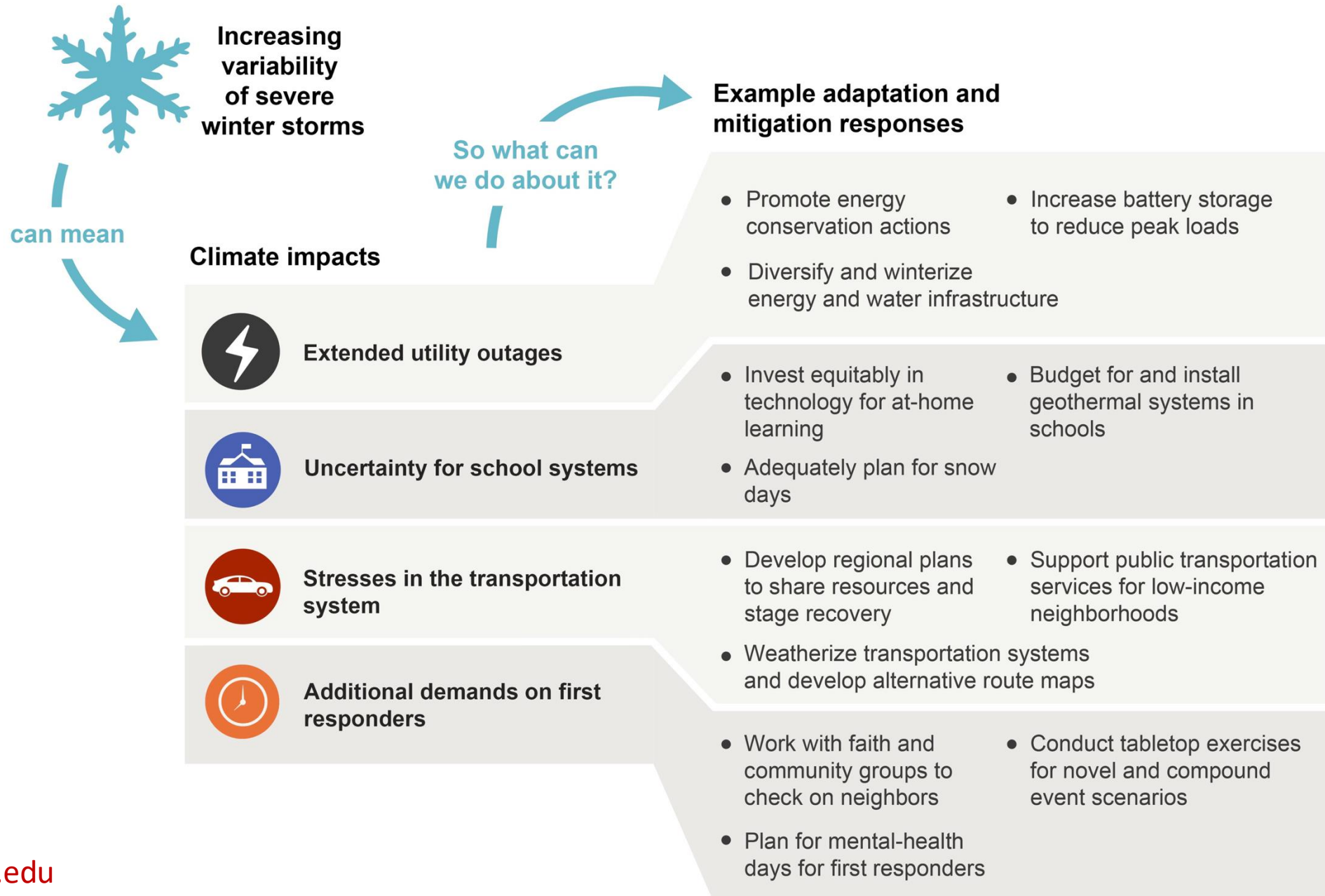
Resilience Actions to Address the Impacts of Hotter Summer Temperatures on Outdoor Activities



Resilience Actions to Address Equity and Justice Issues Related to Increasing Hurricane Risk



Resilience Actions to Address the Impacts of Severe Winter Storms on Public Services



STARTLEMENT

by Ada Limón, 24th Poet Laureate Consultant in Poetry at the Library of Congress

It is a forgotten pleasure, the pleasure
of the unexpected blue-bellied lizard

skittering off his sun spot rock, the flicker
of an unknown bird by the bus stop.

To think, perhaps, we are not distinguishable
and therefore no loneliness can exist here.

Species to species in the same blue air, smoke—
wing flutter buzzing, a car horn coming.

So many unknown languages, to think we have
only honored this strange human tongue.

If you sit by the riverside, you see a culmination
of all things upstream. We know now,

we were never at the circle's center, instead
all around us something is living or trying to live.

The world says, What we are becoming, we are
becoming together.

The world says, One type of dream has ended
and another has just begun.

The world says, Once we were separate,
and now we must move in unison.



Ada Limón, 24th US Poet Laureate



Presentation available at:
www.github.com/SmithEcophysLab/seminar/2025_hsc
nick.smith@ttu.edu



Thanks!